

INTERNSHIP TECHNICAL REPORT

AI Job Market Analysis

Data Cleaning & Exploratory Data Analysis of 5,773 AI-Related Job Postings

Prepared by

Sarfaraz Manzoor Ahmed Korejo

AI / Full Stack Intern — SEIC (Sapphire Consulting Services)

Project: Week 1 — AI Job Market

Report Date: August 12, 2026

Data Analysis Task — No Machine Learning Model Deployed

1. Executive Summary

This report documents a data cleaning and exploratory data analysis (EDA) task on 5,773 AI-related job postings, performed in Python (pandas, NumPy, Matplotlib, Seaborn) as part of AI/Data Analytics skill-building ahead of the SEIC internship.

What was done: The raw dataset was checked for missing values and invalid entries, cleaned via a reusable pipeline, and exported to processed/ai_job_market_clean.csv. EDA then examined job-title demand, geography, experience mix, work arrangement, skills, and salary.

Dataset: 5,773 postings across 5 countries (US, UK, Canada, Germany, Australia), 13 columns, sourced from Adzuna and USAJobs.

Major findings: Data Scientist and AI Engineer are the top titles; the US leads in postings and salary; 82% of postings don't specify a work arrangement; Python, Computer Vision, and NLP are the top requested skills; salaries vary widely by country and show inconsistent experience-level labeling.

Machine learning model: Not applicable — this was a data analysis task; no predictive model was built.

Key result: A clean, analysis-ready dataset with documented demand, skills, and salary insights.

2. Task, Objective & Dataset

Task: Week 1 — AI Job Market: Data Cleaning & Exploratory Data Analysis (Data Analysis task; no ML model).

Context: Not explicitly documented in the project files; inferred from findings.md that the goal was to understand demand, skills, geography, and compensation in the AI job market.

Objective: Clean the dataset and answer: which titles/countries/skills lead demand, what share of jobs are remote, and how salary varies by experience and country.

Dataset: 5,773 rows × 13 columns. No target variable (descriptive analysis only). Covers 2,152 companies and 1,159 cities across 5 countries; posted_date spans 2021–2026.

Feature	Type	Role
job_title, company, city	Categorical / text	Input
country, currency	Categorical	Input
salary_min, salary_max	Numerical	Input
remote_type	Categorical	Input
experience_level	Categorical	Input
required_skills	Text list	Input
posted_date	Date/time	Input

Feature	Type	Role
source	Categorical	Input
job_description	Free text	Input (not mined)

3. Data Quality & Preprocessing

Issues Found

Column	Missing	%
salary_min	2,430	42.1%
salary_max	2,415	41.8%
required_skills	4,689	81.2%
posted_date	123	2.1%
company	5	0.1%

- No duplicate rows found.
- 3 postings had \$0 salary (both min and max) — set to missing.
- 4 postings had salary_min > salary_max (invalid range, e.g. \$160K min vs \$18K max) — set to missing.
- 10 postings had an implausible salary midpoint below \$10,000 — excluded from salary analysis (not yet persisted to the saved cleaned file — see Limitations).

Cleaning Steps Applied

Missing values: company → "Unknown" (5 rows); required_skills → "Not provided" (4,689 rows) — placeholders preserve the row without implying a false value.

Date parsing: posted_date converted to proper datetime using errors="coerce", so unparseable dates become missing rather than breaking the conversion.

Invalid salaries: Zero-value and inverted-range salaries were set to missing rather than guessed at, since there was no reliable way to infer the correct value.

Output: Cleaned dataset (5,773 rows retained, no rows dropped — only invalid values nulled out) saved to processed/ai_job_market_clean.csv.

4. Methodology

The workflow followed a standard analysis sequence: (1) load the raw CSV and inspect shape, dtypes, and a preview; (2) quantify missing values and identify invalid salary entries; (3) apply the targeted cleaning fixes above and persist the cleaned CSV; (4) run univariate and grouped exploratory analysis — frequency counts by job title, country, experience level, and remote type, plus skill-frequency extraction and salary aggregation; (5) visualize each result with Matplotlib/Seaborn. The pipeline is reproducible: rerunning

cleaning_pipeline.py followed by the EDA notebook on the same raw CSV reproduces every figure in this report.

5. Exploratory Data Analysis

5.1 Job Titles & Geography

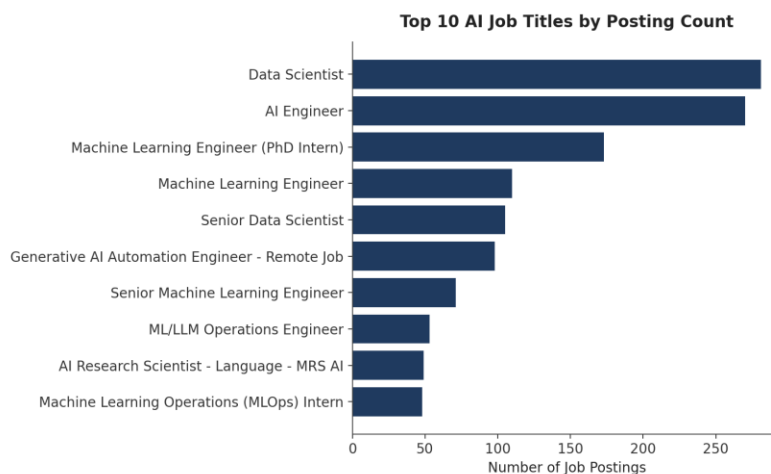


Figure 1. Top 10 AI job titles by posting count.

Data Scientist (281) and AI Engineer (270) lead, followed by Machine Learning Engineer (PhD Intern) (173) and Machine Learning Engineer (110) — demand is concentrated in a handful of core roles.

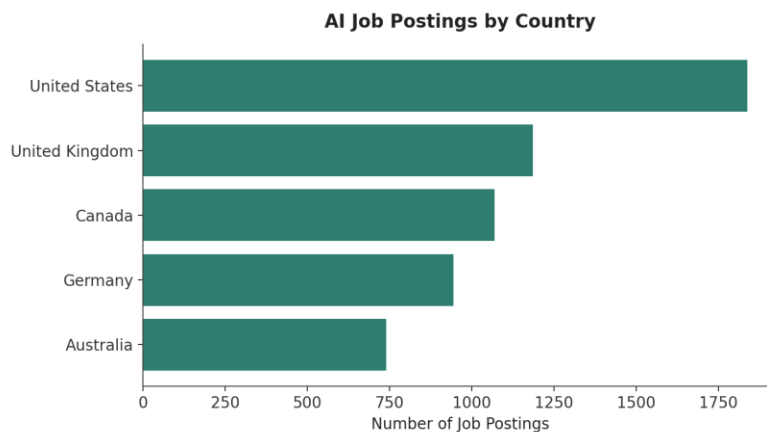


Figure 2. Postings by country.

US: 1,837 (31.8%), UK: 1,185 (20.5%), Canada: 1,068 (18.5%), Germany: 944 (16.4%), Australia: 739 (12.8%) — a fairly even 5-country spread, not US-dominated.

5.2 Experience Level & Work Arrangement

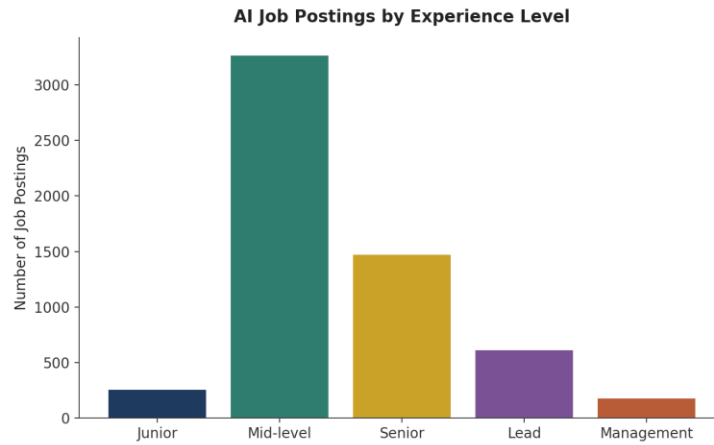


Figure 3. Postings by experience level.

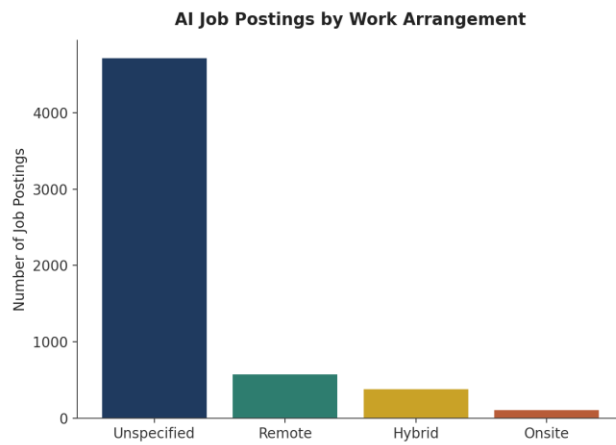


Figure 4. Postings by work arrangement.

Mid-level roles dominate (56.5%); Junior postings are just 4.4% of the dataset. 81.7% of postings don't specify a work arrangement, so remote-work share cannot be reliably estimated from this data.

5.3 Required Skills

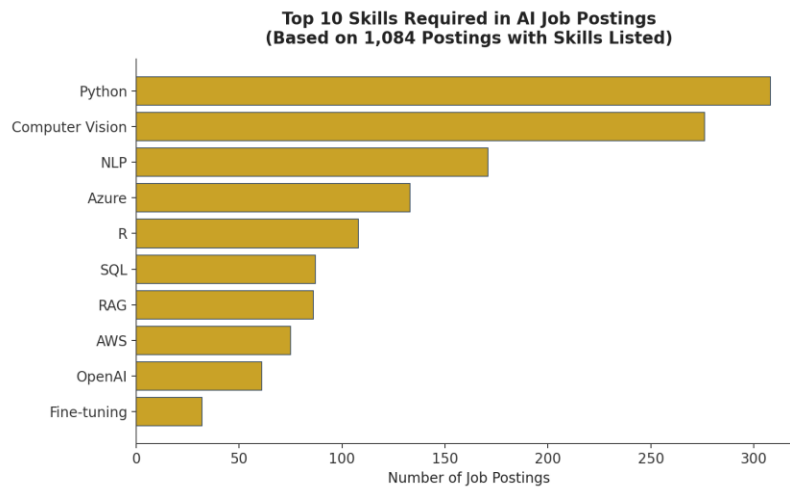


Figure 5. Top 10 skills (based on the 1,084 postings, 18.8%, that listed skills).

Python (308) and Computer Vision (276) lead, followed by NLP (171), Azure (133), R (108), SQL (87), RAG (86), AWS (75), OpenAI (61), Fine-tuning (32). Cloud and generative-AI/LLM skills (RAG, OpenAI, Fine-tuning) feature prominently alongside core Python/CV/NLP skills. This ranking is directional only, since 81% of postings list no skills at all.

5.4 Salary Analysis

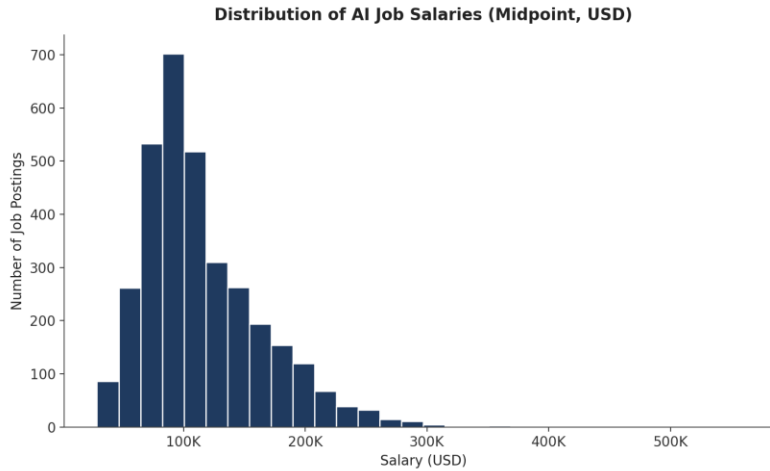


Figure 6. Salary distribution (midpoint, USD; outliers <\$10K excluded).

Based on 3,322 postings, median salary is \$103,000 (mean \$115,294; range \$29K–\$565K). The right-skewed distribution (mean > median) reflects a smaller number of high-paying postings.

Country	n	Median	Mean
United States	1,821	\$120,000	\$132,610
Australia	64	\$107,500	\$109,789
Canada	200	\$100,000	\$105,905
United Kingdom	1,180	\$82,000	\$91,458
Germany	57	\$56,500	\$94,667

Experience Level	n	Median	Mean
Junior	137	\$125,000	\$130,657
Mid-level	2,061	\$97,000	\$107,068
Senior	665	\$122,000	\$131,483
Lead	370	\$111,500	\$125,385
Management	89	\$109,000	\$119,247

US and UK have the largest, most reliable samples; Germany and Australia are small (57–64 postings) and less robust. Notably, median salary does not rise steadily with seniority — Junior (\$125K) and Senior (\$122K) outrank Mid-level (\$97K). Closer inspection shows this is driven by experience_level mislabeling: several top-paid "Junior" postings carry clearly senior titles (e.g. "NLP Engineer – Sr. Associate" at \$361,000), so with

only 137 Junior postings, a few mislabeled records skew the median. This is a data-quality finding, not a genuine market pattern.

6. Key Findings

- Demand is concentrated in Data Scientist, AI Engineer, and ML Engineer roles across a balanced 5-country spread led by the US.
- 81.7% of postings omit work arrangement, so remote-work prevalence can't be reliably estimated.
- Python, Computer Vision, and NLP lead requested skills (among the 18.8% of postings listing skills); generative-AI/LLM skills (RAG, OpenAI, Fine-tuning) are emerging strongly.
- Median salary is \$103,000 overall, ranging from \$56,500 (Germany) to \$120,000 (US).
- The `experience_level` field shows labeling inconsistencies that distort salary-by-seniority comparisons — job title is a more reliable seniority signal in some cases.

7. Limitations

- Skill ranking is based on only 18.8% of postings (skills missing in 81.2%).
- Salary findings cover ~58% of postings with usable data; disclosure may not be random.
- Experience-level labeling inconsistencies affect seniority-based salary comparisons.
- Germany and Australia salary samples are small (57–69 postings) — treat as indicative only.
- The $< \$10,000$ outlier filter was applied only during EDA, not saved back to the cleaned CSV.
- No free-text/NLP analysis of `job_description`; no statistical significance testing performed.

8. Recommendations & Conclusion

- Use skill and remote-work findings as directional guidance only, given the large gaps in those fields.
- Benchmark compensation using medians (not means), weighted toward the larger US/UK samples.
- Cross-check `experience_level` against job title before using it for seniority-based analysis.
- Persist the low-salary outlier filter into `cleaning_pipeline.py` so the saved CSV is fully salary-clean.
- Future work: NLP-based skill extraction from `job_description`, correcting experience-level labels, and adding statistical testing to country/seniority comparisons.

This task successfully turned a raw, partially inconsistent dataset of 5,773 AI job postings into a clean, analysis-ready dataset with documented findings on demand, skills, and compensation. The objective — understanding this AI job market segment — was achieved within the scope and limitations noted above. No ML model was built, as none was required for this task.

9. Technical Environment

Python, pandas 3.0.5, NumPy 2.5.2, Matplotlib 3.11.1, Seaborn 0.13.2, run in Jupyter Notebook (EDA_AI_Job_Market.ipynb), per requirements.txt. No database or external API used — all I/O via local CSV files. To reproduce: run `cleaning_pipeline.py` on `data/ai_job_market.csv`, then execute the EDA notebook top-to-bottom.